LAND USE LAND COVER CLASSIFICATION

A Remote Sensing & GIS Analysis of Jhelum District, Punjab, Pakistan

Using Sentinel-2 Multispectral Imagery, ESA WorldCover Reference Data,
and a Random Forest Classifier in Google Earth Engine

Prepared by

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GIS & Remote Sensing Portfolio — Project 1 of 15
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1. Executive Summary

This report documents the design, implementation, and validation of a Land Use Land Cover (LULC) classification for Jhelum District, Punjab, Pakistan, developed entirely in Google Earth Engine (GEE) using Sentinel-2 Level-2A surface reflectance imagery. The study classifies the district into five land cover categories — Water, Vegetation, Agriculture, Bare Soil, and Built-up — using a Random Forest classifier trained on a hybrid dataset combining automated stratified sampling from ESA WorldCover v200 with manually digitized reference polygons.

The classification was refined through several iterative stages: incorporation of a second, wet-season image composite to exploit seasonal vegetation change; addition of a spectral brightness index; and inclusion of a GLCM texture (contrast) feature to resolve spectral confusion between Bare Soil and Built-up surfaces. The final model achieved an overall accuracy of 77.2% (Kappa = 0.71) against an independent, district-wide reference dataset, and 92.4% (Kappa = 0.90) against manually validated, spatially independent ground-truth polygons.

Results were exported, quantified by area (km²) and percentage of district coverage, and compiled into a cartographically finished map layout in QGIS, including a Pakistan-wide locator inset. This document summarizes the objectives, data, methodology, results, challenges encountered, and their solutions.

2. Project Objectives

- Classify Jhelum District into five land cover classes using multispectral satellite imagery.
- Apply a supervised machine learning approach (Random Forest) suitable for professional/operational remote sensing workflows.
- Develop and justify a training data strategy in the absence of formal field survey data.
- Conduct a rigorous, methodologically sound accuracy assessment — including an independent validation set free of spatial autocorrelation bias.
- Diagnose and iteratively resolve sources of classification error using established remote sensing techniques.
- Quantify land cover extent by area and percentage, and produce a publication-quality cartographic output.

3. Study Area

Jhelum District is located in northern Punjab, Pakistan, bounded by the Jhelum River, and characterized by a mix of urban settlement (Jhelum city and Sohawa), riverine and canal-fed agriculture, remnant forest/scrub vegetation, and seasonally fallow cropland. The district's total mapped area was computed as 3,616.12 km², based on the classification output itself.

4. Data Sources

Dataset	Provider / Platform	Purpose
Sentinel-2 Level-2A (Surface Reflectance, Harmonized)	Copernicus / ESA, via Google Earth Engine	Primary multispectral imagery (dry season Jan–May 2024; wet season Jul–Sep 2024)
FAO GAUL Level 2 Administrative Boundaries	FAO, via Google Earth Engine	Study area (district) and country boundary delineation
ESA WorldCover v200	European Space Agency	Reclassified as automated training/reference labels
Manually digitized polygons	Author-created in GEE Code Editor	Independent training and validation samples

5. Methodology

5.1 Image Acquisition and Preprocessing

Sentinel-2 imagery was filtered to the study area and date range, with cloud and cirrus contamination removed using the QA60 bitmask band. A cloud-free median composite was generated for the dry season to serve as the primary classification input, and a second composite for the wet/monsoon season was generated to capture seasonal vegetation dynamics. Exact acquisition windows and filtering parameters for each season are given below.

Parameter	Dry Season Composite	Wet Season Composite
Date range	1 Jan 2024 - 31 May 2024 (5 months)	1 Jul 2024 - 30 Sep 2024 (3 months)
Scenes used (post-filter)	46 scenes	Filtered separately per cloud threshold below
Max cloud cover pre-filter	5% (CLOUDY_PIXEL_PERCENTAGE)	30% (CLOUDY_PIXEL_PERCENTAGE)
Cloud/cirrus pixel masking	Applied (QA60 bitmask)	Applied (identical QA60 bitmask function)
Compositing method	Per-pixel median across all filtered scenes	Per-pixel median across all filtered scenes

Note on cloud filtering thresholds: both seasonal composites used the identical pixel-level cloud/cirrus masking function; only the scene-level pre-filter threshold differed. This was a deliberate, climate-informed decision rather than an inconsistency. Punjab's monsoon season (July-September) is inherently far cloudier than the Jan-May dry season; applying the same strict 5% threshold to wet-season imagery would likely have returned few or no usable scenes. The threshold was relaxed to 30% for the wet season specifically to ensure enough scenes were available to build a valid, complete composite, while the pixel-level masking function still removed cloud/cirrus-contaminated pixels from every scene used, regardless of that scene's overall cloud percentage.

5.2 Spectral Indices and Derived Features

In addition to six raw spectral bands (Blue, Green, Red, NIR, SWIR1, SWIR2), the following derived features were computed to improve class separability:

- NDVI (Normalized Difference Vegetation Index) — vegetation vigor.
- NDWI (Normalized Difference Water Index) — surface water detection.
- NDBI (Normalized Difference Built-up Index) — built-up surface indicator (noted limitation: confusable with dry/bare soil in single-season imagery).
- Brightness Index — mean visible-band reflectance.
- Wet-season NDVI and seasonal NDVI difference (wet minus dry) — captures the growing-season greening signature unique to cultivated cropland.
- GLCM Texture (Contrast) — spatial heterogeneity measure distinguishing structurally uniform surfaces (bare soil, cropland) from heterogeneous built environments (roads, rooftops, shadow).

5.3 Training Data Strategy

In the absence of field survey data, a hybrid training strategy was adopted, combining two independent sources:

- Automated stratified samples (2,500 points per class; 12,500 total) drawn from ESA WorldCover v200, reclassified from its native 11 classes into the study's 5-class schema.

- Manually digitized training polygons (5 classes, drawn from visual interpretation of true-color and false-color composites), pixel-sampled and capped at 2,500 pixels per class to prevent one data source from dominating the model.
- A second, spatially separate set of manually digitized validation polygons was drawn specifically to avoid spatial autocorrelation bias (i.e., testing on pixels adjacent to training pixels), providing an independent accuracy check.

Combining both sources balanced the strengths of each: the automated samples provided consistent, whole-district coverage, while the manual polygons provided cleaner, expert-verified examples of each class.

5.4 Classification

A Random Forest classifier (300 decision trees) was trained on the combined dataset using thirteen input features: six raw spectral bands, three spectral indices, brightness, wet-season NDVI, seasonal NDVI difference, and GLCM texture contrast.

5.5 Accuracy Assessment

Two independent accuracy assessments were conducted to provide a balanced, methodologically honest evaluation:

- (A) Automated Test Set — a 30% held-out split of the WorldCover-derived stratified samples, representing whole-district generalization performance.
- (B) Independent Manual Validation — a spatially separate set of hand-labeled validation polygons, representing performance on unambiguous, expert-verified land cover examples.

5.5.1 Note on Compositing Window Length

A multi-month acquisition window (5 months for the dry season, 3 months for the wet season) was used to build each seasonal composite, rather than a single-month window. This is a deliberate methodological choice with genuine trade-offs, summarized below, rather than a single objectively 'correct' approach.

Approach	Advantage	Risk / Limitation
Narrow window (1 month)	Captures a precise temporal snapshot; all pixels represent a similar crop growth stage or point in the season, minimizing temporal blending.	Sentinel-2's ~5-day revisit interval, combined with cloud filtering, may leave very few clear observations in a single month - risking incomplete, gapped composites, particularly during the monsoon season.
Wide window (multi-month, used here)	Guarantees sufficient clear observations exist to build a complete, gap-free composite across the full study area, improving reliability without field verification.	Blends imagery from slightly different points within the season (e.g., early- vs. late-stage crop growth), which can somewhat soften the sharpness of the seasonal signal.

Neither approach is universally superior; the choice depends on whether temporal precision or spatial completeness is prioritized. A wider window was selected for this project to prioritize a complete, gap-free composite over the district's full extent.

5.6 Iterative Refinement

Initial classification using only raw bands and standard indices showed pronounced confusion between Agriculture, Bare Soil, and Built-up classes — a known limitation attributable to spectral similarity of dry/fallow surfaces in single-season imagery. Two targeted refinements were introduced sequentially and evaluated against both accuracy metrics:

- Wet-season NDVI + seasonal NDVI difference — added to exploit the fact that active cropland greens during the monsoon season while true bare soil and built-up surfaces do not. This measurably improved Agriculture classification accuracy.
- GLCM texture (contrast) — added to exploit the structural difference between uniform bare surfaces and heterogeneous built environments. This produced the largest single improvement in the project, reducing Bare Soil↔Built-up misclassification by 37–70% across both accuracy checks.

6. Results

6.1 Final Accuracy Metrics

Assessment	Overall Accuracy	Kappa Coefficient	Interpretation
(A) Automated Test Set (whole-district, independent reference)	77.15%	0.7144	Substantial agreement
(B) Independent Manual Validation (expert-verified samples)	92.35%	0.9009	Almost perfect agreement

The gap between these two figures is expected and methodologically meaningful: (A) reflects performance across the full complexity of the district, including mixed and transitional pixels, while (B) reflects performance on visually unambiguous reference examples. Reporting both — rather than a single number — provides a more complete and defensible picture of model performance than either figure alone.

6.2 Effect of Texture Feature on Class Confusion

Confusion Pair	Before Texture	After Texture	Reduction
Bare Soil → Built-up (Automated)	195	123	-37%
Built-up → Bare Soil (Automated)	57	31	-46%
Bare Soil → Built-up (Manual Validation)	648	375	-42%
Built-up → Bare Soil (Manual Validation)	141	42	-70%
Agriculture → Built-up (Manual Validation)	325	10	-97%

6.3 Land Cover Area Statistics

Final class areas were computed at native 10 m pixel resolution across the full study area (3,616.12 km^2):

Class	Area (km^2)	% of District
Water	124.13	3.43%
Vegetation	1,304.63	36.08%
Agriculture	1,357.48	37.54%
Bare Soil	522.32	14.44%
Built-up	307.56	8.51%

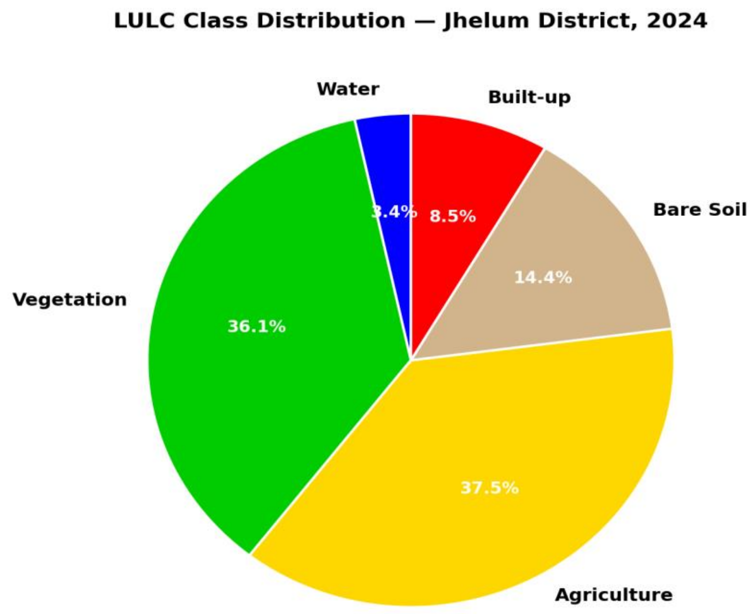


Figure 1. Proportional land cover distribution, Jhelum District, 2024.

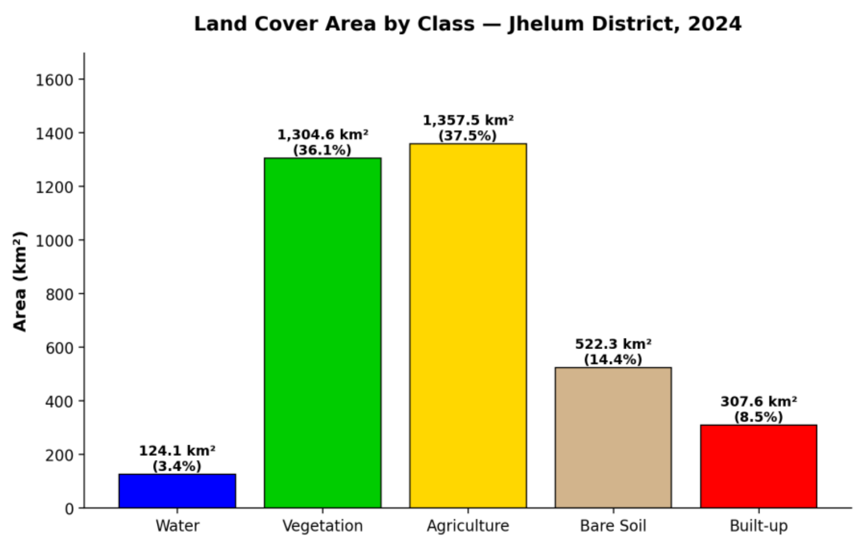
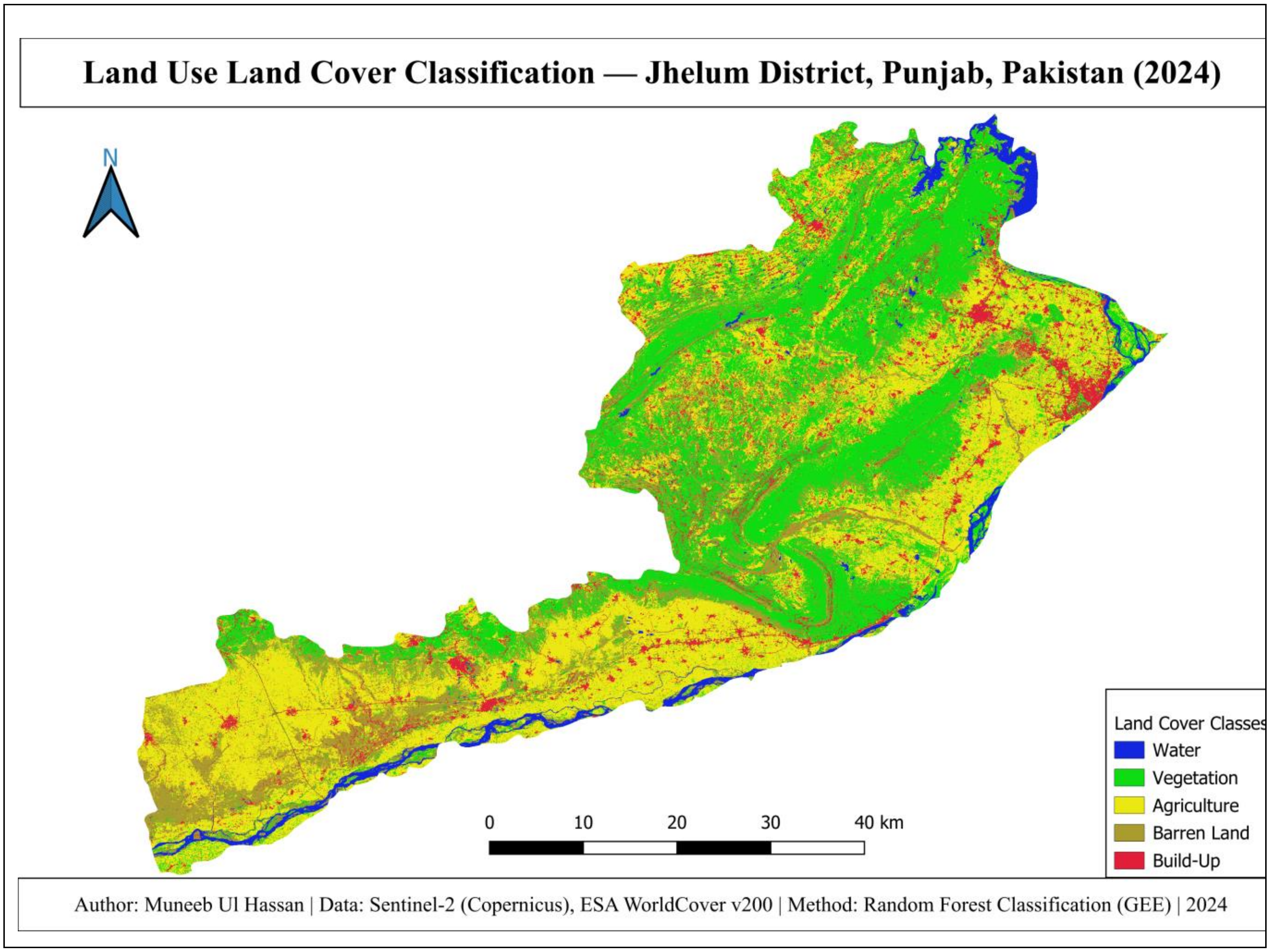


Figure 2. Land cover area by class (km²), Jhelum District, 2024.

Agriculture and Vegetation together account for over 73% of the district's land area, consistent with Jhelum's predominantly rural, riverine agricultural character. Built-up area (8.51%) is concentrated around Jhelum city and secondary towns such as Sohawa.

6.4 Final Classification Map



7. Challenges and Solutions

Challenge	Solution
No field survey / ground-truth data available	Adopted a hybrid training strategy combining automated ESA WorldCover-derived samples with manually digitized, expert-verified polygons.
Risk of inflated accuracy from spatial autocorrelation (training/testing pixels drawn from the same polygons)	Digitized a second, spatially separate set of validation polygons, and reported both an automated and an independent accuracy figure rather than a single number.
NDBI misclassifying dry/fallow cropland as built-up	Diagnosed via confusion matrix inspection; addressed by adding a wet-season NDVI seasonal-change feature to distinguish active cropland from permanently bare/built surfaces.
Persistent Bare Soil ↔ Built-up spectral confusion	Introduced a GLCM texture (contrast) feature to capture the structural difference between uniform bare surfaces and heterogeneous built environments — the single largest accuracy improvement in the project.

Challenge	Solution
Manual polygon pixel counts (100,000+) overwhelming automated sample counts (~4,000) in the training set	Applied per-class capping/subsampling to balance contributions from both data sources.
Interactive Console timeouts when computing district-wide area statistics at full 10 m resolution	Migrated the computation to a Google Earth Engine batch Export task (with tileScale optimization), which runs without the interactive time limit while preserving full spatial resolution and accuracy.
Shapefile export failure due to mixed geometry types inherited from source boundary dataset	Applied a geometry dissolve operation prior to export to guarantee a single, valid geometry type.

8. Cartographic Output

The final classification raster was imported into QGIS for cartographic finishing. The map layout includes: a paletted symbology matching the five defined land cover classes; a true-color composite for visual context; the district administrative boundary; a legend, scale bar, and north arrow; and a Pakistan-wide locator inset map with an automatically generated extent-indicator rectangle showing Jhelum District's location within the country.

9. Conclusion and Skills Demonstrated

This project demonstrates an end-to-end remote sensing and GIS workflow: satellite image acquisition and preprocessing, spectral feature engineering, supervised machine learning classification, rigorous and methodologically honest accuracy assessment, iterative error diagnosis and resolution, spatial statistics, and professional cartographic output. The dual-metric accuracy reporting approach and texture-based error correction in particular reflect applied problem-solving beyond a standard tutorial-level classification.

Skills demonstrated:

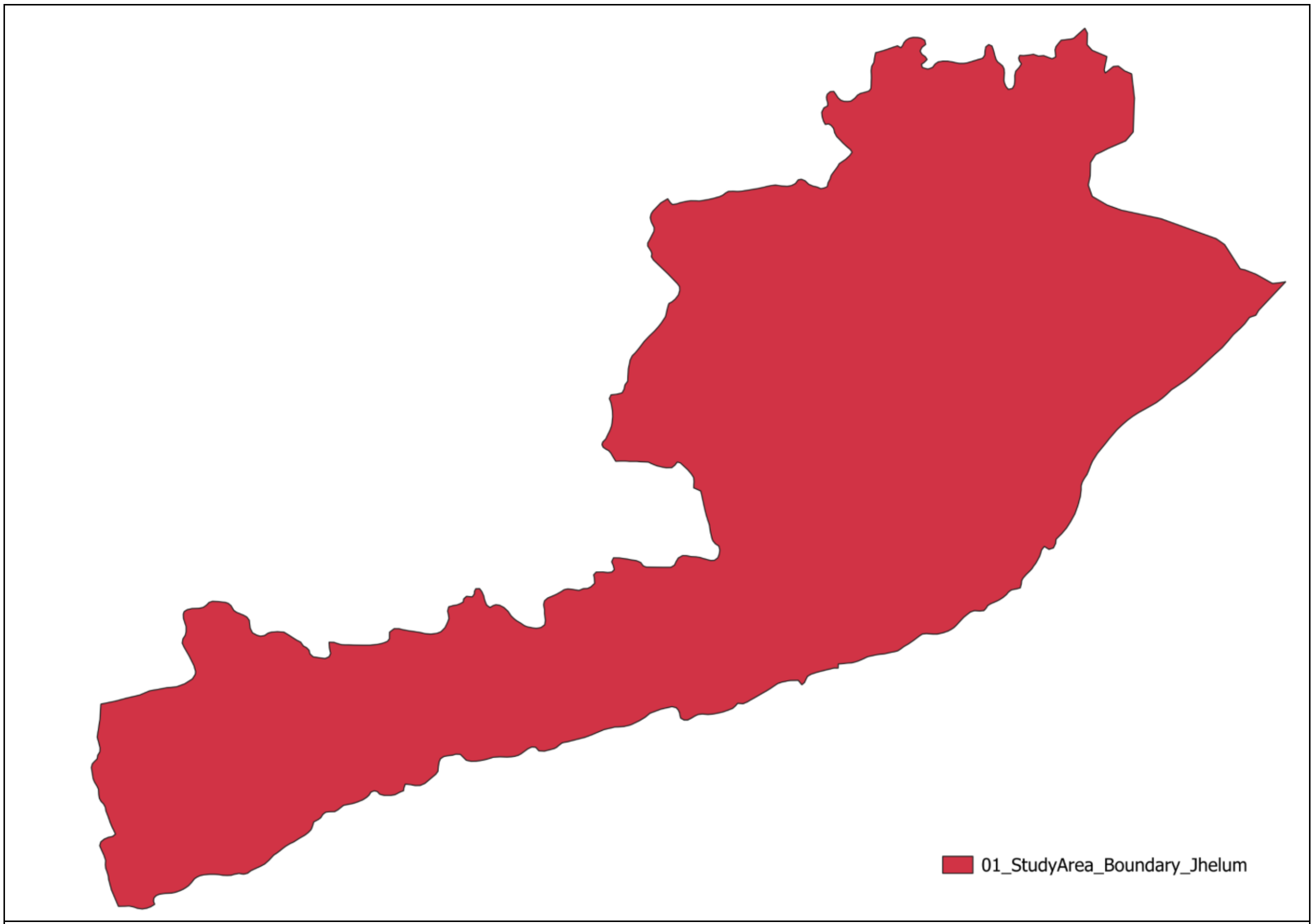
- Google Earth Engine (JavaScript API) — image collection filtering, cloud masking, compositing, classification, batch export.
- Multispectral remote sensing — spectral index derivation (NDVI, NDWI, NDBI), seasonal composite analysis, GLCM texture analysis.
- Machine learning for Earth observation — Random Forest classification, stratified sampling, feature engineering.
- Accuracy assessment methodology — confusion matrix analysis, Kappa statistics, spatial autocorrelation awareness.
- QGIS cartography — symbology, print layout composition, locator/inset mapping.
- Spatial statistics — area computation and land cover proportion analysis.

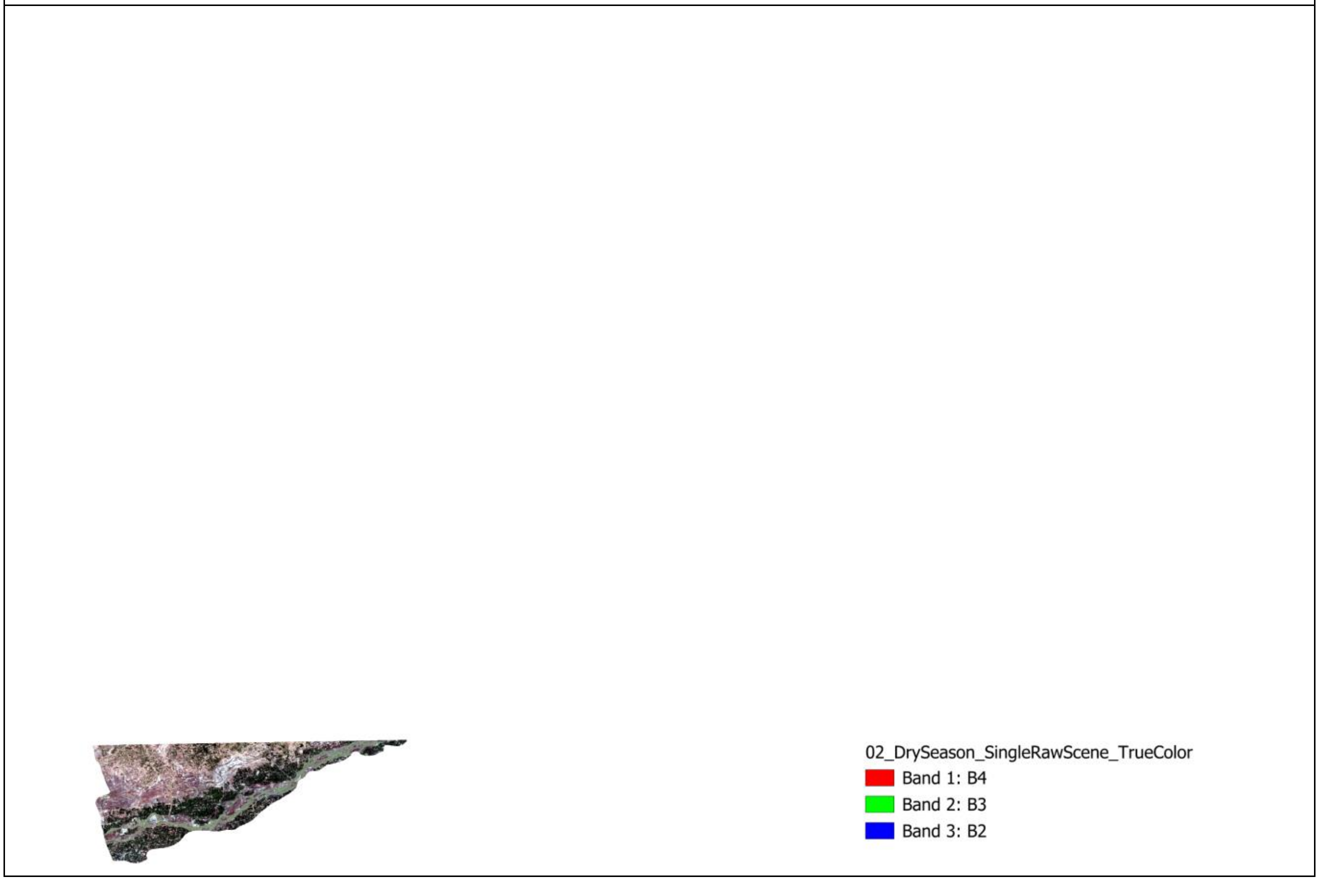
Appendix A: Supplementary Layer Gallery

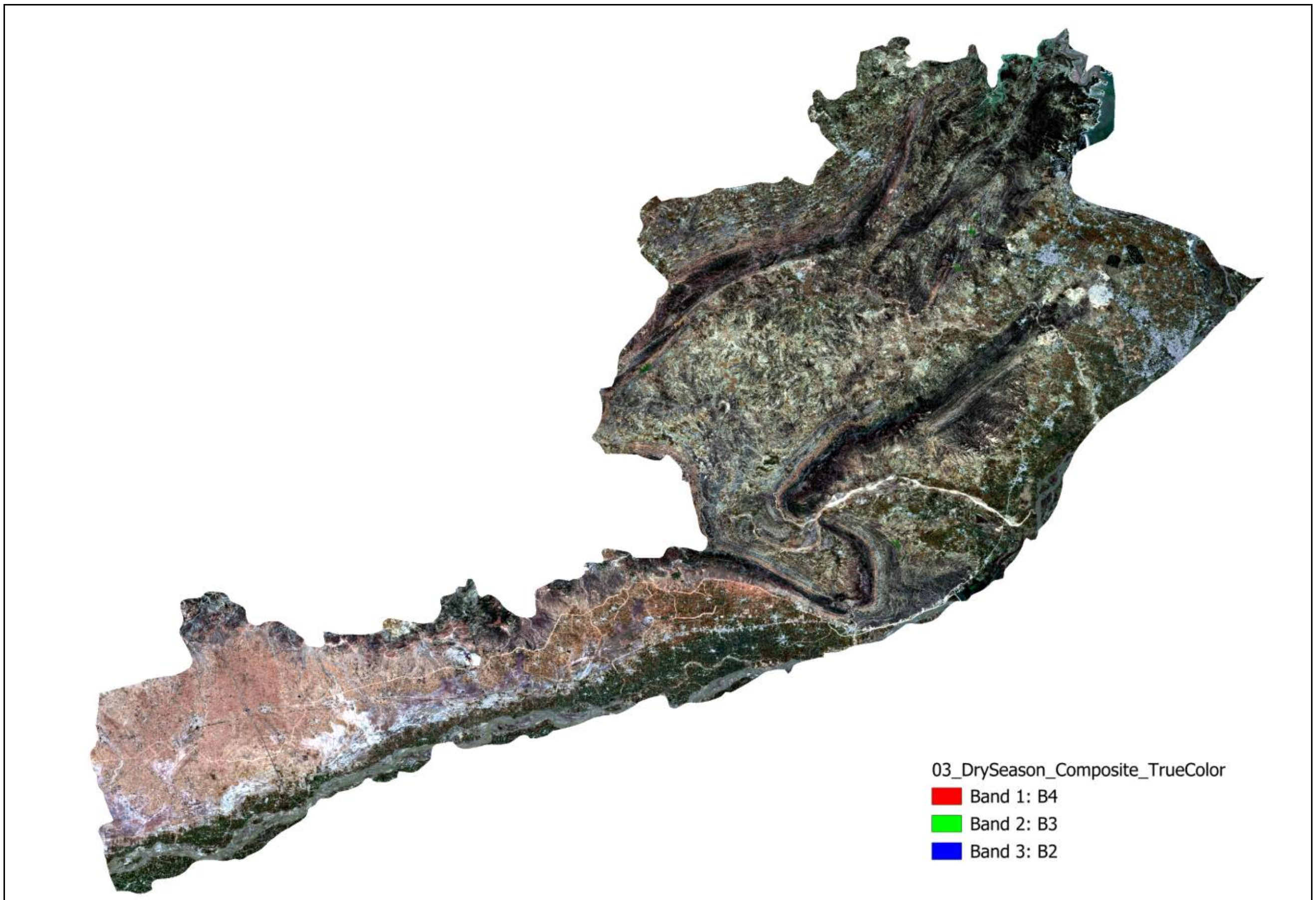
The classification workflow produced sixteen distinct outputs (imagery, indices, reference, and result layers), each displayed and inspected individually in Google Earth Engine before being combined for classification. Full metadata for each is listed below; corresponding exported files should be inserted as figures in the marked positions.

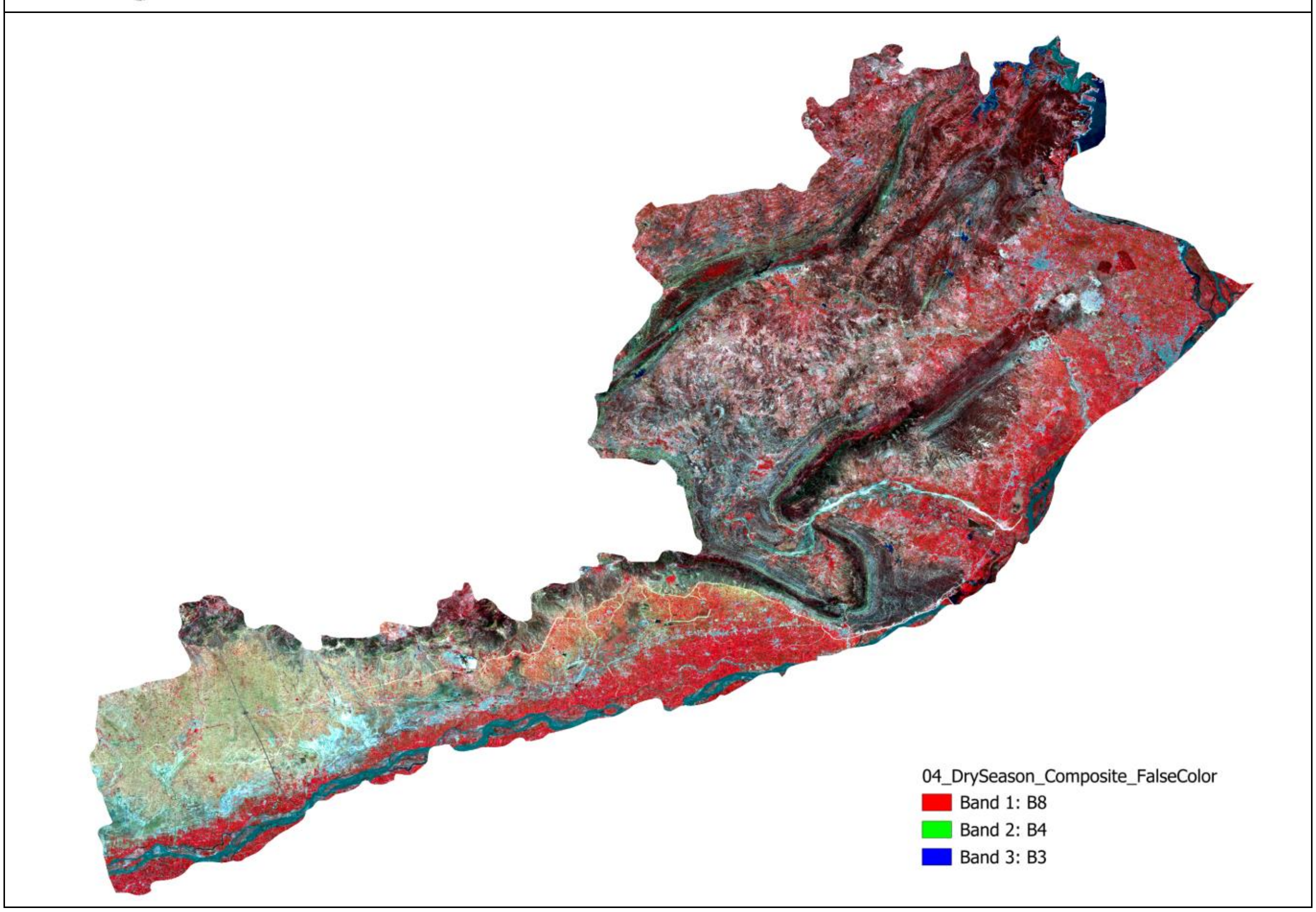
A.1 Study Area and Source Imagery

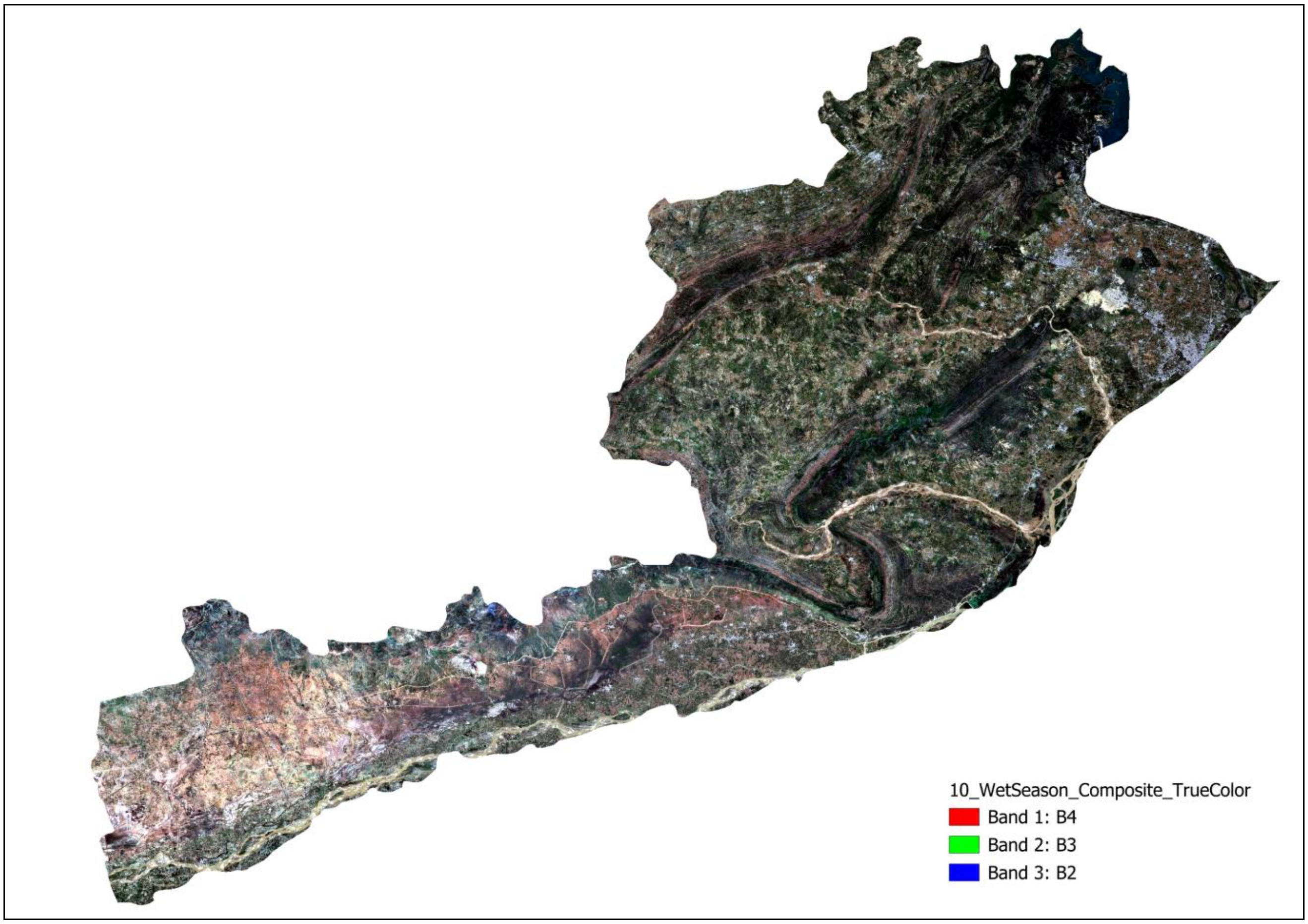
#	Layer	Date Range	Description
01	Study Area Boundary	N/A (FAO GAUL 2015)	Jhelum District administrative boundary, used to clip all subsequent layers.
02	Dry Season - Single Raw Scene (Uncomposited)	First clear scene, Jan-May 2024	One individual Sentinel-2 acquisition prior to compositing; shown to illustrate residual gaps/artifacts a single date can contain.
03	Dry Season - Composite True Color	1 Jan - 31 May 2024	Cloud-free median composite, bands B4/B3/B2 (Red/Green/Blue), natural-color view.
04	Dry Season - Composite False Color	1 Jan - 31 May 2024	Bands B8/B4/B3 (NIR/Red/Green); healthy vegetation renders bright red.
10	Wet Season - Composite True Color	1 Jul - 30 Sep 2024	Cloud-free median composite for the monsoon growing season, natural-color view.


01_StudyArea_Boundary_Jhelum


02_DrySeason_SingleRawScene_TrueColor
Band 1: B4
Band 2: B3
Band 3: B2


03_DrySeason_Composite_TrueColor
Band 1: B4
Band 2: B3
Band 3: B2

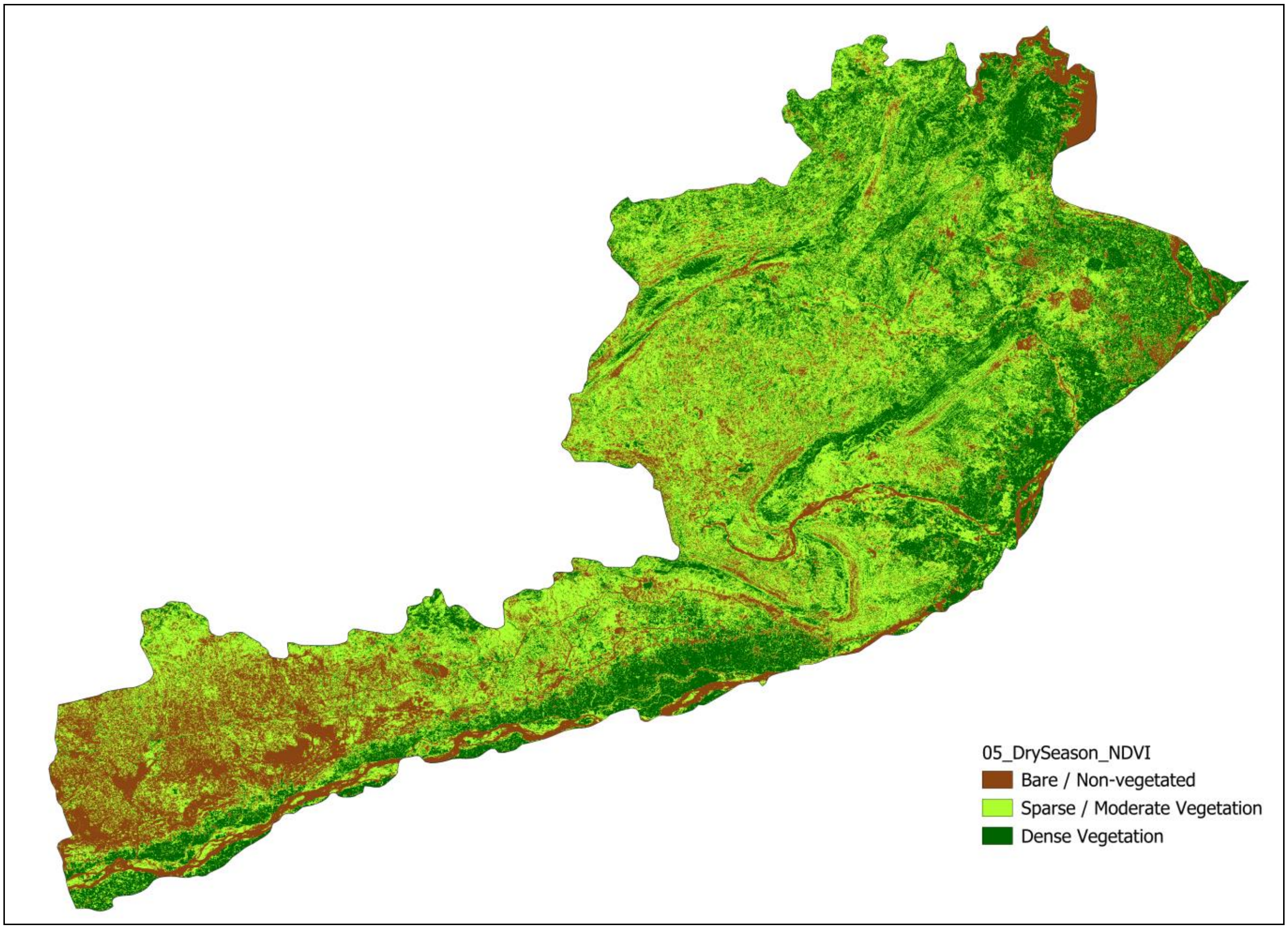

04_DrySeason_Composite_FalseColor
Band 1: B8
Band 2: B4
Band 3: B3

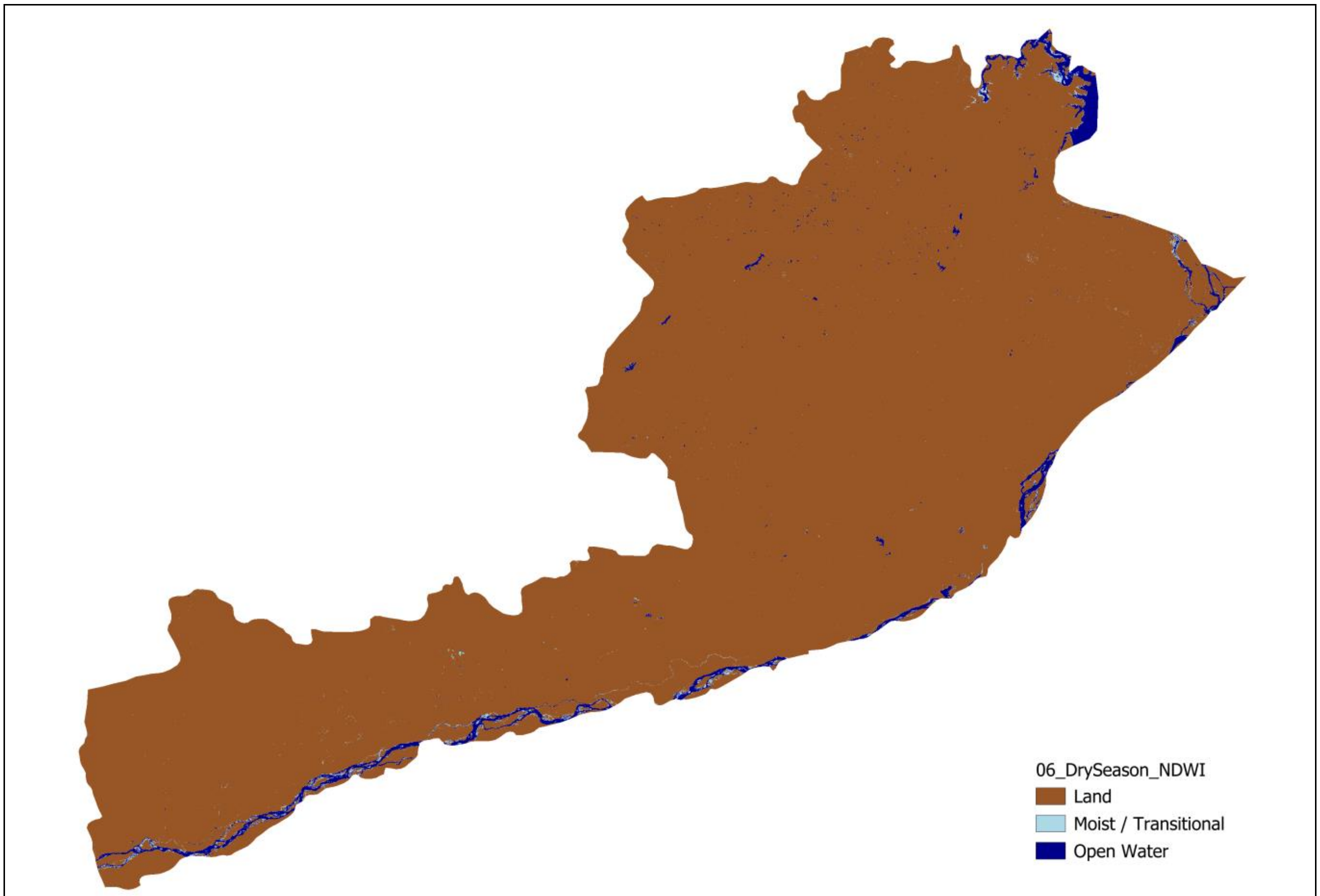


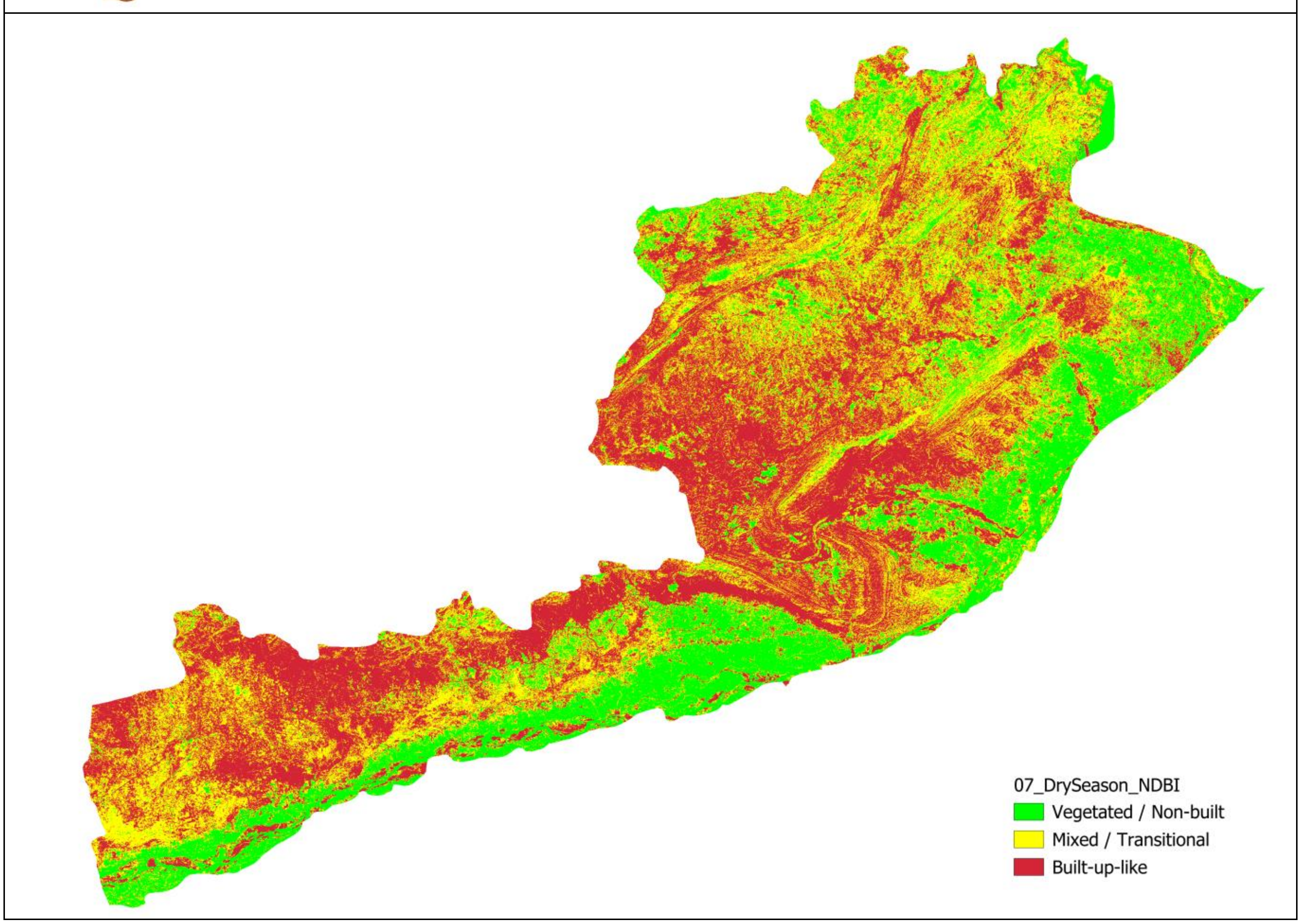
A.2 Spectral Indices and Derived Features

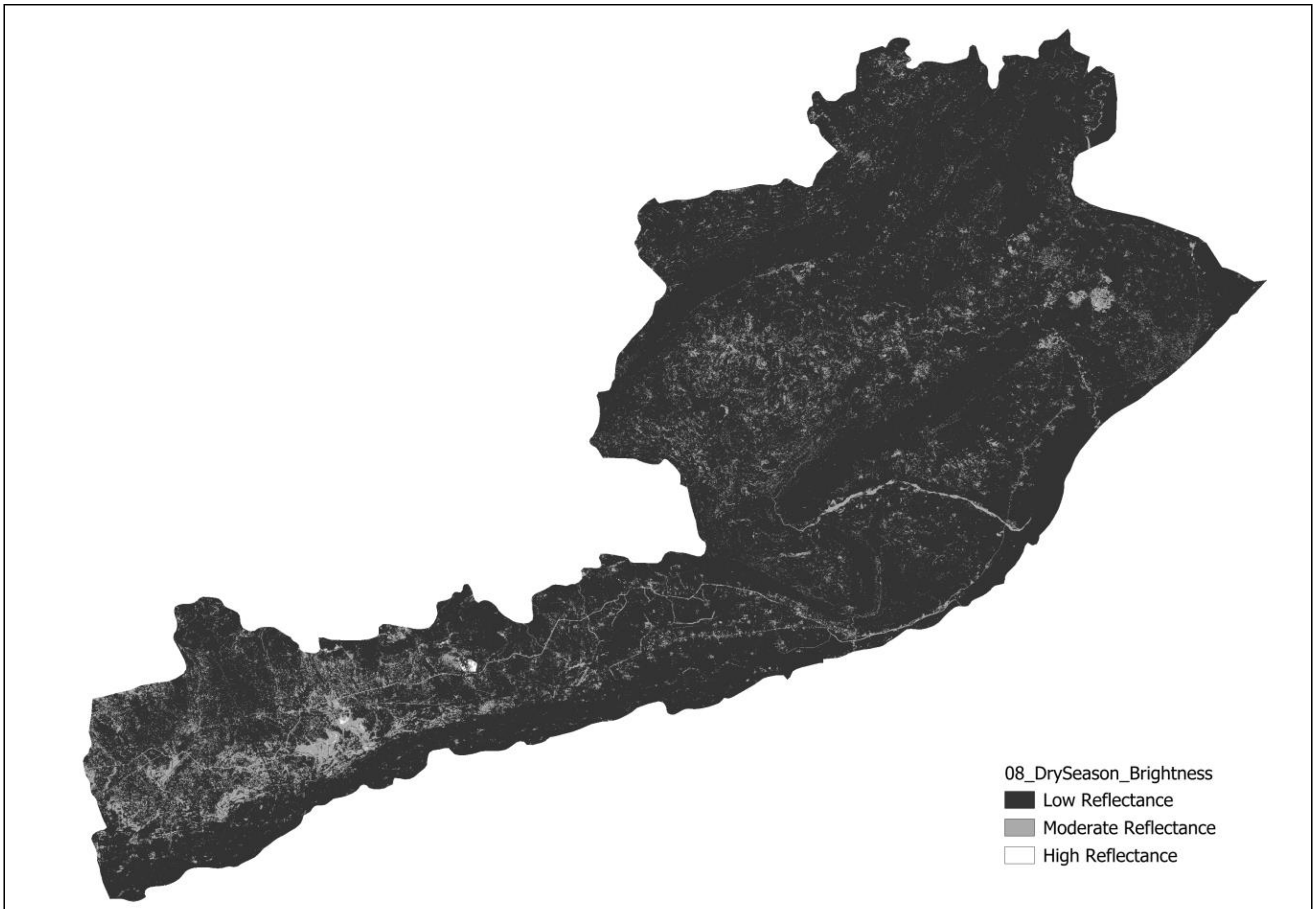
#	Layer	Season / Date Range	Description
05	NDVI - Dry Season	Dry: 1 Jan - 31 May 2024	(NIR-Red)/(NIR+Red). High values indicate dense, healthy vegetation.
06	NDWI - Dry Season	Dry: 1 Jan - 31 May 2024	(Green-NIR)/(Green+NIR). High values indicate open water / high moisture content.
07	NDBI - Dry Season	Dry: 1 Jan - 31 May 2024	(SWIR-NIR)/(SWIR+NIR). High values suggest built-up surfaces, but also confusable with dry/bare soil in single-season imagery (see Section 5.6).
08	Brightness Index	Dry: 1 Jan - 31 May 2024	Mean reflectance across B2/B3/B4. Used to help separate built-up from vegetated surfaces; found insufficient alone to separate built-up from bare soil.
11	NDVI - Wet Season	Wet: 1 Jul - 30 Sep 2024	Same NDVI formula applied to the monsoon composite; captures peak-growing-season vegetation vigor.

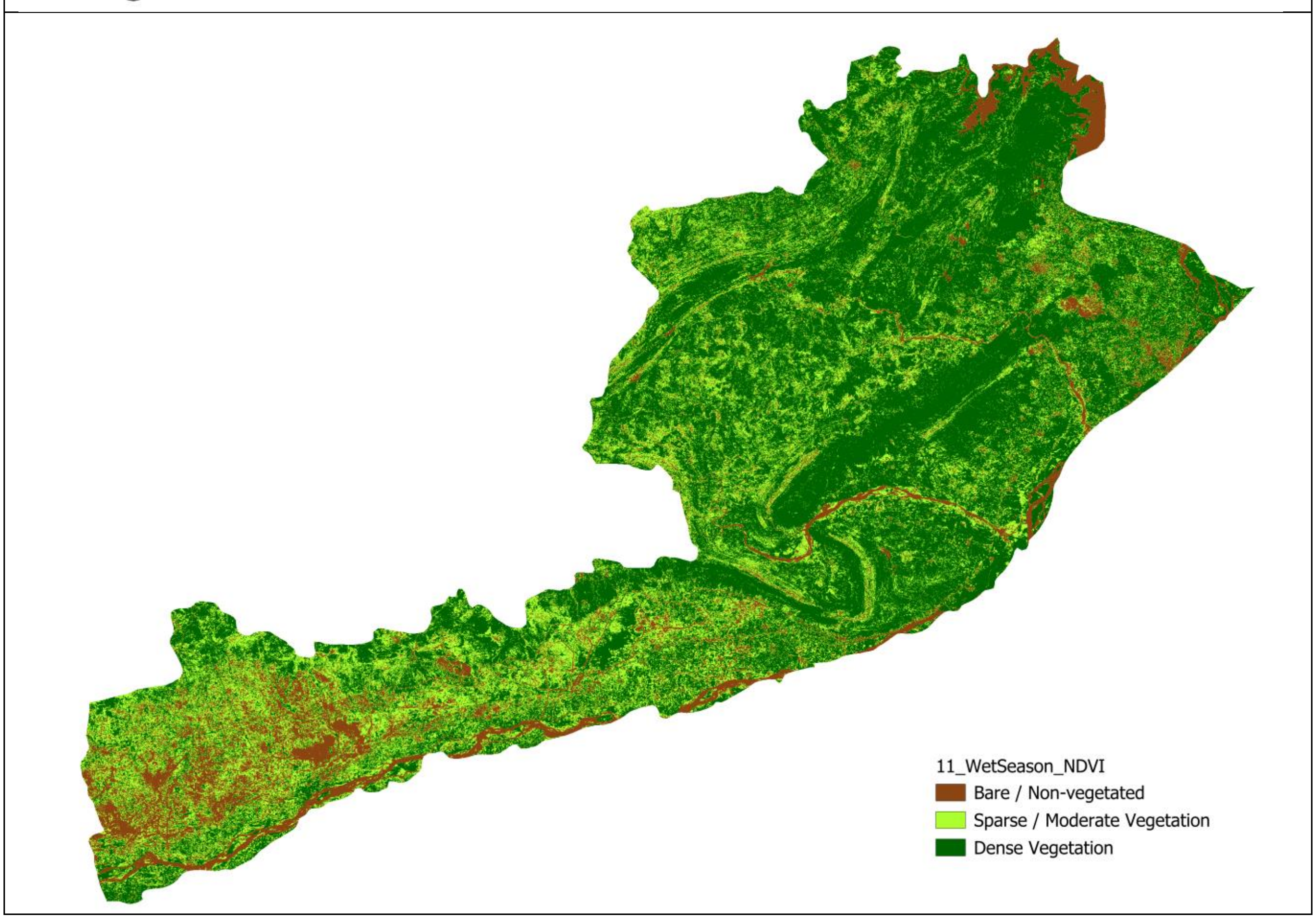
#	Layer	Season / Date Range	Description
12	NDVI Seasonal Change (Wet minus Dry)	Derived from both seasons	Wet-season NDVI minus dry-season NDVI. Active cropland shows a strong positive value (greens up between seasons); true bare soil/built-up remain near zero in both seasons.
09	GLCM Texture (Contrast)	Dry: 1 Jan - 31 May 2024	Spatial contrast measure over a 3x3 window on the brightness band. Distinguishes spatially uniform surfaces (bare soil, fields) from spatially heterogeneous ones (built-up).

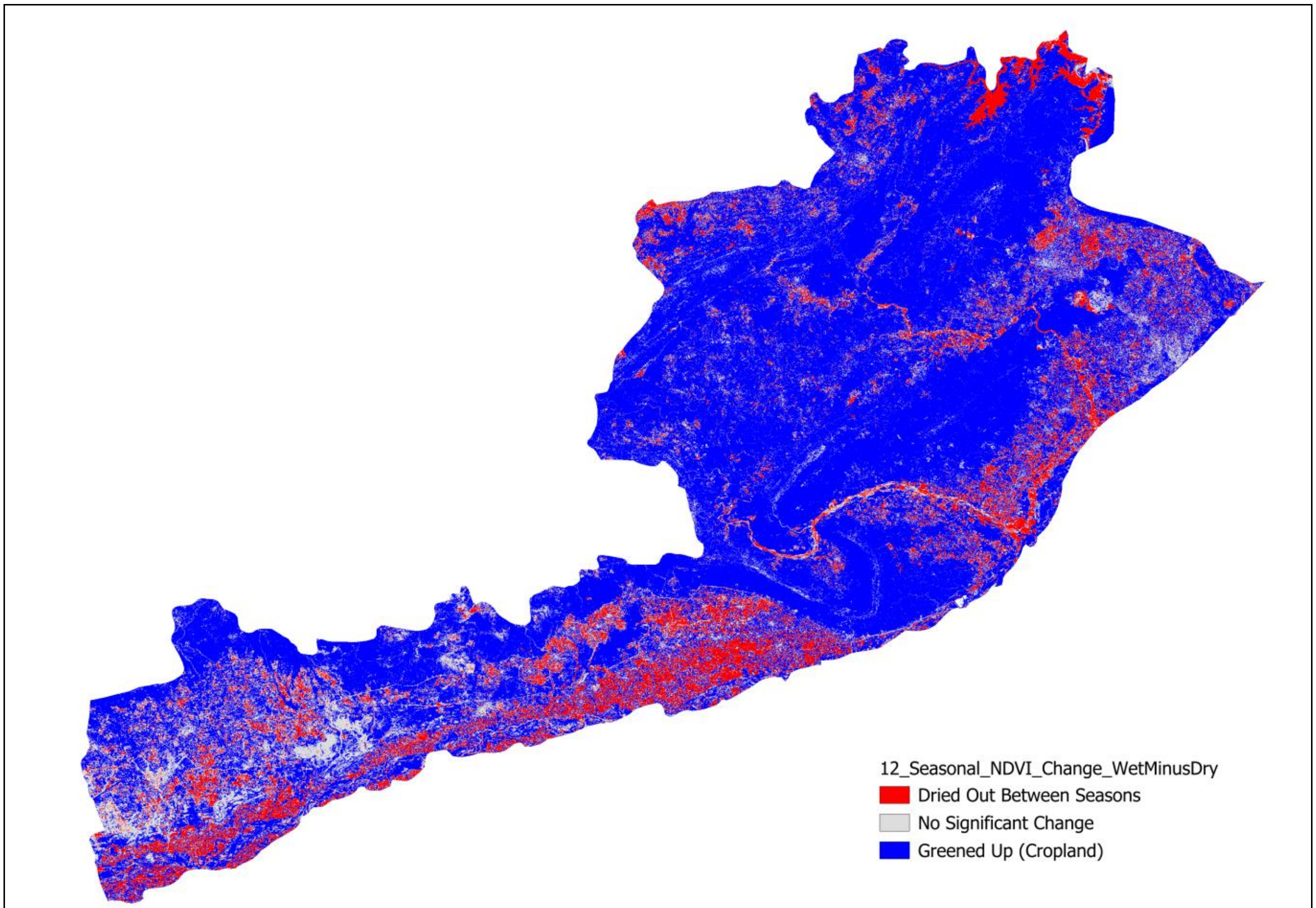


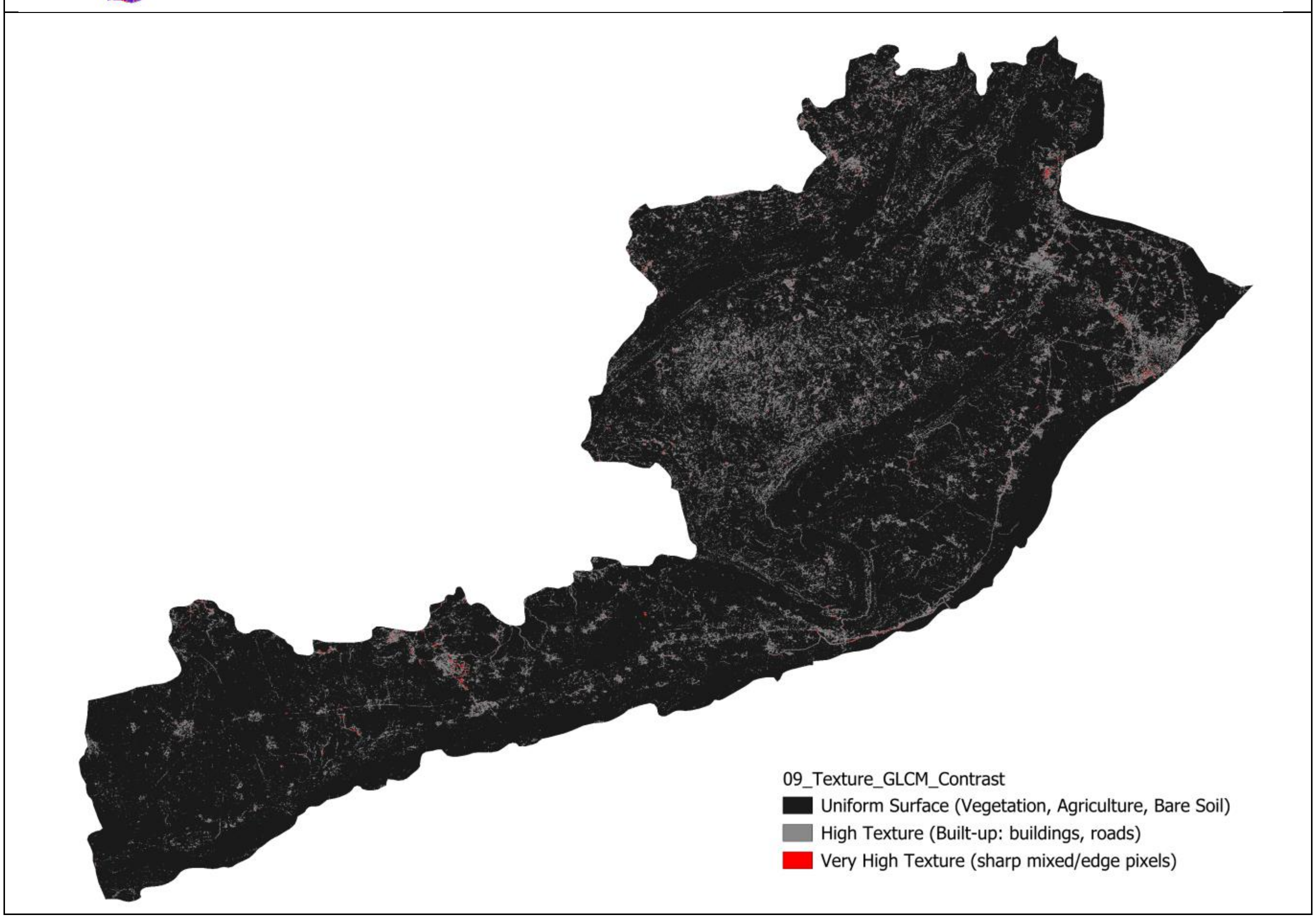

06_DrySeason_NDWI
Land
Moist / Transitional
Open Water


07_DrySeason_NDBI
Vegetated / Non-built
Mixed / Transitional
Built-up-like


08_DrySeason_Brightness
Low Reflectance
Moderate Reflectance
High Reflectance

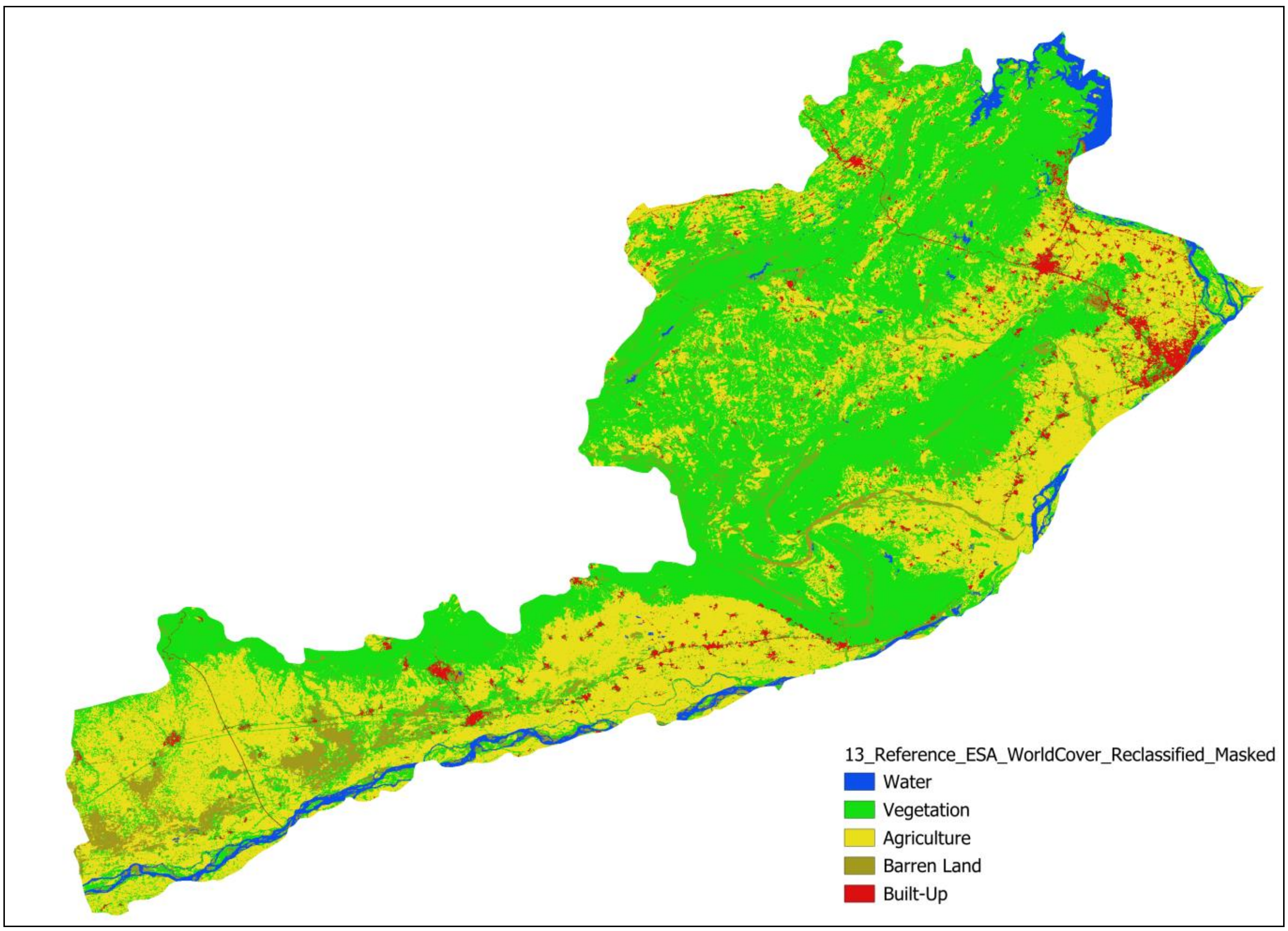

11_WetSeason_NDVI
Bare / Non-vegetated
Sparse / Moderate Vegetation
Dense Vegetation

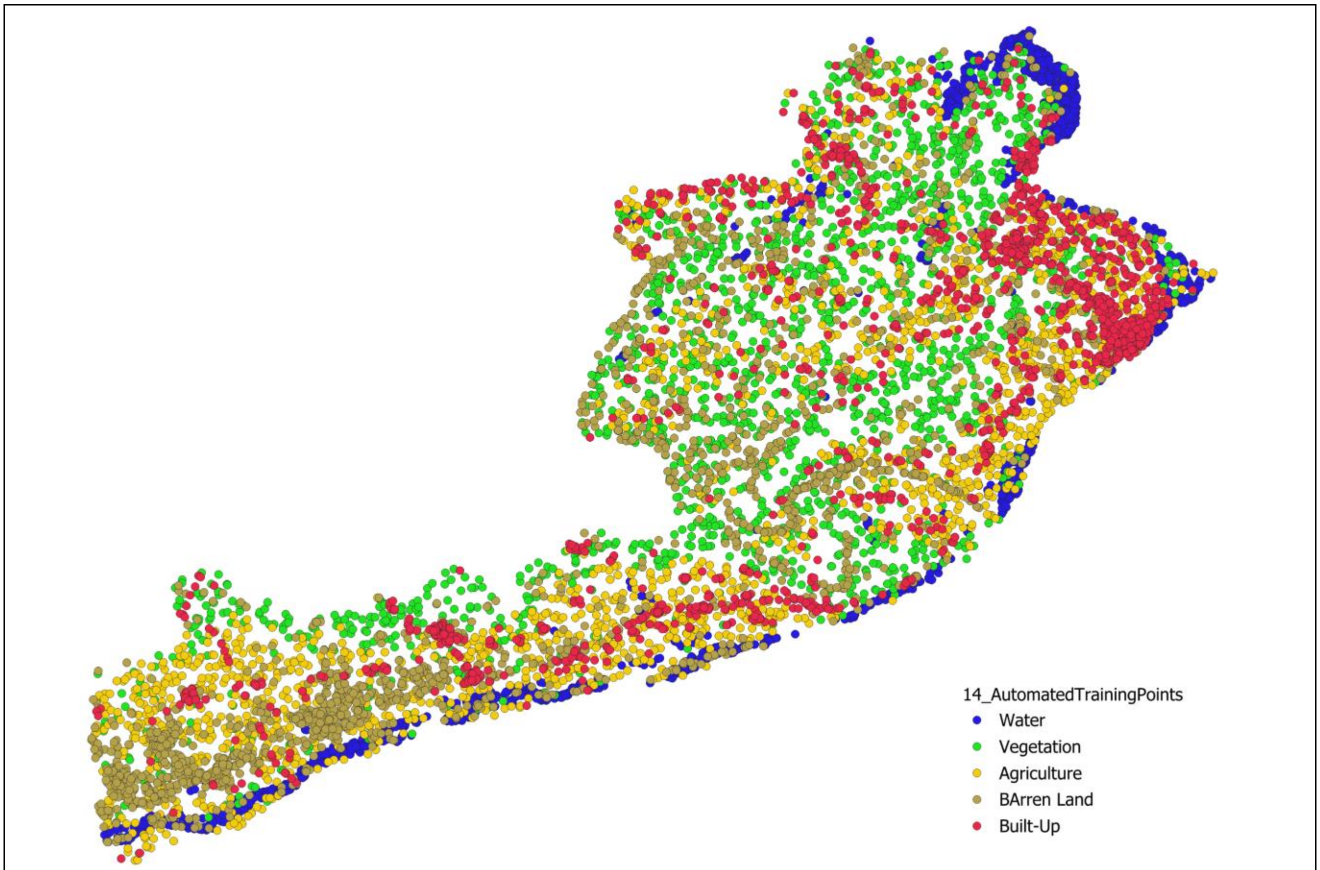

12_Seasonal_NDVI_Change_WetMinusDry
Dried Out Between Seasons
No Significant Change
Greened Up (Cropland)


09_Texture_GLCM_Contrast
Uniform Surface (Vegetation, Agriculture, Bare Soil)
High Texture (Built-up: buildings, roads)
Very High Texture (sharp mixed/edge pixels)

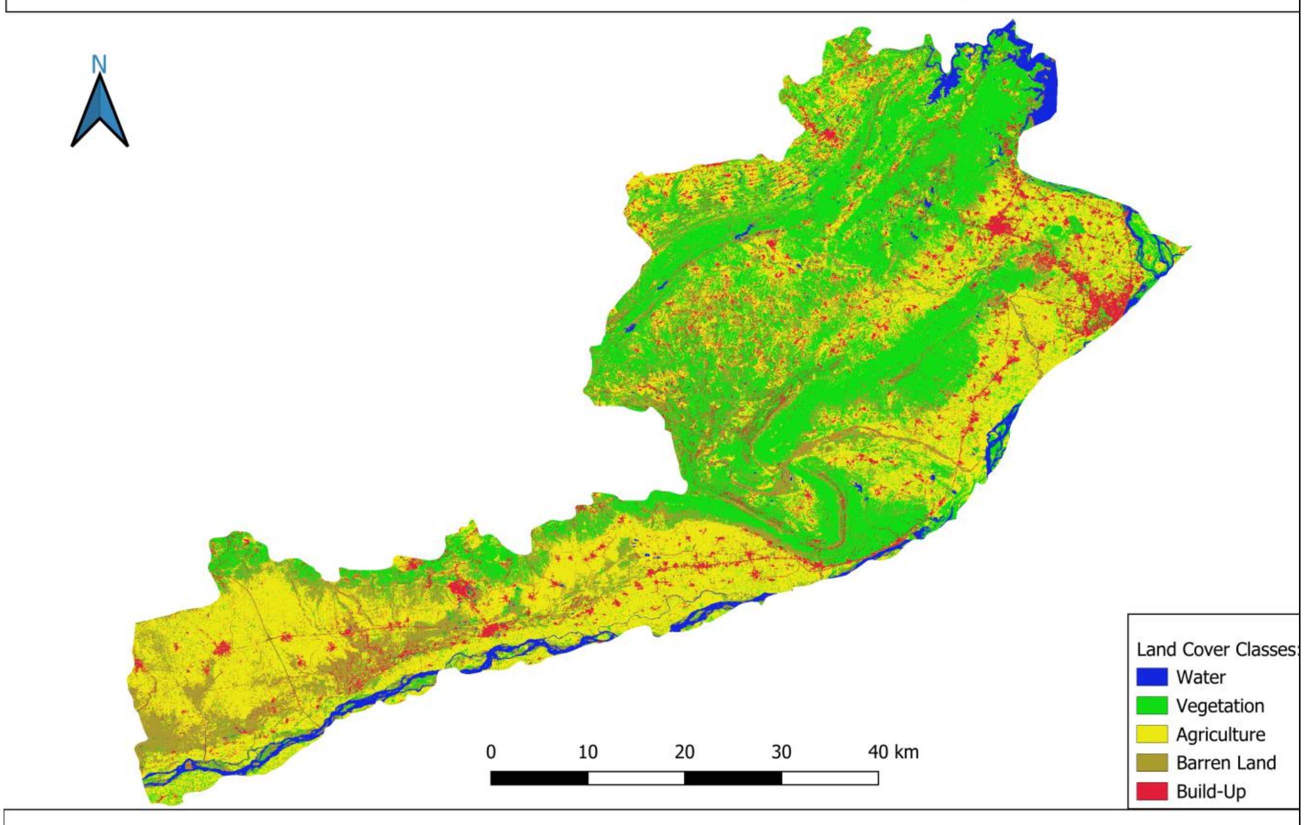
A.3 Reference and Result Layers

#	Layer	Source / Date	Description
13	Reference Land Cover	ESA WorldCover v200 (2021 baseline)	Global 10 m land cover product, reclassified from 11 native classes into the study's 5-class schema; used as the source for automated training samples.
14	Automated Training Points	Derived from WorldCover, 2024 imagery	6,000 (later 12,500) stratified sample points color-coded by class, shown over the study area.
15	Final LULC Classification	Random Forest output, 2024	The final 5-class classified map: Water, Vegetation, Agriculture, Bare Soil, Built-up.





Land Use Land Cover Classification — Jhelum District, Punjab, Pakistan (2024)



Author: Muneeb Ul Hassan | Data: Sentinel-2 (Copernicus), ESA WorldCover v200 | Method: Random Forest Classification (GEE) | 2024

10. Tools and Technologies

Category	Tools / Technologies
Cloud Processing & Analysis	Google Earth Engine (JavaScript API)
Data Sources	Sentinel-2 (Copernicus/ESA), ESA WorldCover v200, FAO GAUL
Machine Learning	Random Forest (smileRandomForest, GEE)
Desktop GIS / Cartography	QGIS
Output Formats	GeoTIFF, Shapefile, CSV, PNG, PDF

Author: Muneeb Ul Hassan | Project 1 of 15, GIS & Remote Sensing Self-Study Portfolio | 2026